

Shivam Jadhav

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SUMMARY

Innovative and results-driven Software Development Engineer with hands-on experience in developing cutting-edge AI and ML solutions. I specialize in designing scalable, distributed systems and algorithms to address complex challenges. With expertise in Python and C++. My focus is on creating transformative technologies that not only improve customer experiences but also contribute to the advancement of AI at scale. Eager to contribute to Amazon's mission of shaping the future of AI and ML.

EDUCATION & RESEARCH

Master's in Computer Science | University of North Texas, Denton, USA

Expected May 2025

- **Relevant Coursework:** Software Engineering, Special Problems, Artificial Intelligence, Computer Architecture
- **Research: Telemetry Data Simulation** – Designed a secure IoT communication framework using microservices, TCP/IP, socket programming, and raw data handling for seamless data exchange with a central server.

AI & MACHINE LEARNING SKILLS

- **NLP & Text Summarization:** Built a Topic-Aware Text Summarization model using language models; excelled in NLP coursework (A grade).
- **Generative AI & LLMs:** Trained and fine-tuned Stable Diffusion, NanoGPT with Hugging Face, PyTorch, Transformers.
- **AI Infrastructure:** Configured Red Hat servers, deployed NVIDIA drivers, CUDA, CUDA Toolkit, and managed on-prem AI infrastructure.
- **Deep Learning & Research:** Implemented GANs, transformers, attention models by applying research-driven methodologies in NLP & Computer Vision.
- **Tech Stack:** Proficient in PyTorch, NumPy, Pandas, Seaborn, Hugging Face for AI development and visualization.
- **AI Software Development:** Optimized AI workflows, trained NanoGPT, and deployed models for real-world applications.
- **MERN Stack:** Built real-time, scalable applications using MongoDB, Express.js, React.js (Hooks), and Node.js, serving 10,000+ users.
- **Cloud DevOps:** Managed AWS infrastructure (EC2, Lambda, DynamoDB, Route 53, S3, API Gateway, CloudWatch). Skilled in Linux, Bash scripting, and system services (Unit files).
- **Systems Servers:** Expertise in Windows/Linux, server imaging, RAID setup, NVIDIA AI server configuration, and shell scripting.

RESEARCH

AI Bug Priority Problem | Research | MS-CS

Aug 2024 – Dec 2024 | Denton Texas, USA

- Conducted a research project on AI accessibility under Professor Wajidi Aljedani, focusing on the "Bug Priority Problem."
- Trained a custom model using ALBERT, RoBERTa, and BERT, achieving 90% accuracy with RoBERTa and up to 96% accuracy with ALBERT after extensive data curation.links:-  **AI-Model**  **plugin**

Railway Bogie Defect Detection using AI and IoT

- Implemented a backend service for data curation and defect detection from live data feeds.
- Trained a classification model to detect railway part defects under varying lighting conditions using PyTorch, sci-kit-learn, and Python.
- Developed frontend and backend services for tracking defects and sending alerts across camera feeds, using React, Tailwind CSS, Firebase, and Nodemailer.

EXPERIENCE

IT Specialist - Student Assistant | University of North Texas, Denton, USA

Sep 2023 – Present

- Designed and deployed innovative solutions, including a Universal Digital Signage Product and Raspberry Pi Clusters, enhancing the university's digital infrastructure and reducing operational costs by \$40,000.
- Engineered and automated critical systems, including a status-checking application for monitoring server uptime, which proactively detected downtime and sent real-time alerts to improve service reliability.
- Managed server imaging (Ubuntu 24, RAID), SSH, CUDA installations, and full server rack management for AI research, utilizing Grafana for monitoring.

Full - Stack Engineer | Cyrrup Solutions, Hyderabad, India

Mar 2021 – Aug 2023

- Built full - stack applications for IoT fleet management and led cloud microservices integration for performance and scalability.
- Collaborated with cross - functional teams on high - performance cloud projects, optimizing network systems and performing basic troubleshooting (DNS, TCP/IP).

Freelance Full - Stack Developer | ArkTech Ltd.

Jan 2019 – Apr 2021

- Developed and maintained Ark CRM for financial institutions (MERN), streamlining transactions and client management for 10,000+ users.
- Built and integrated AI - driven news recommendation features using custom web scraping, AI models like summarization and cosine similarity, based on user transaction history.

PROFESSIONAL PROJECTS

Universal Digital Signage System | University of North

Sep 2023 – Present | Denton, Texas, USA

- Built and deployed a scalable digital signage solution with React.js, Node.js, MongoDB, UI/UX, database - design, Docker and Raspberry Pi, saving \$40,000.
- Designed a secure platform with JWT/OAuth and developed custom Raspberry Pi OS for seamless device control.
- Automated device configuration, improving efficiency and reducing manual work.

Vehicle Black Box (Fleet Management and AI Solution)

- Built a fleet management system using Vue.js, Node.js, Redis, Python, and cloud services (AWS, GCP) for high scalability.
- Integrated microservices architecture to manage 40+ IoT devices.

CERTIFICATIONS AWARDS

- Deep Learning
- Meta React Basics
- Meta Front - End Development
- Google: Tools of the Trade: Linux and SQL
- Amazon Web Services Cloud Practitioner
- Meta HTML and CSS in depth
- Programming with Javascript
- Version Control
- Docker for Developers

REFERENCES

Michael Baggett, Director | Assistant Dean of Information Technology

Wajdi Aljedaani UX Human - Centered AI Researcher and Consultant

Gaurav Kumar Inviting Tech and Business for Road Safety Solutions